CYBR 345 Test 3

Instructions:

- Do not delete, edit, or change any text or questions in this document.
- Type or write your answers directly below each question.
- Show all work and clearly explain your reasoning using complete sentences and sufficient detail.
- If submitting handwritten work, ensure images are clear, properly oriented, and easy to read.
- Submit your work only through Blackboard; do not email it.
- Answers that do not show required work or follow instructions will receive a grade penalty or a grade of 0.

Q1. **25 Points**

The following message was encrypted by user **A** using a rotation (Caesar) cipher. The key for the rotation cipher was securely shared using the Diffie-Hellman key exchange.

Only uppercase letters are used, with the following mapping:
A = 0, B = 1, …, W = 22
(Letters X, Y, Z are not used)
The original message is:

THE CAT IS A SECRET AGENT

Given:

- $g = 5, p = 23$
- A chooses private key $a = 2$
- B chooses private key $b = 3$

1. Using the Diffie-Hellman algorithm, what is A's public key?

1) A public key

$A_{pub} = g^a \% p$

$= 5^2 \% 23$

$= 25 \% 23$

$= 2$

$\boxed{A_{pub} = 2}$

2. Using the Diffie-Hellman algorithm, what shared key does A compute using B's public key? Show your calculation.

$B_{pub} = g^b \% p = 125 \% 23 = 10$

3

2) $K = (B_{pub})^a \% p$

$= (10)^2 \% 23$

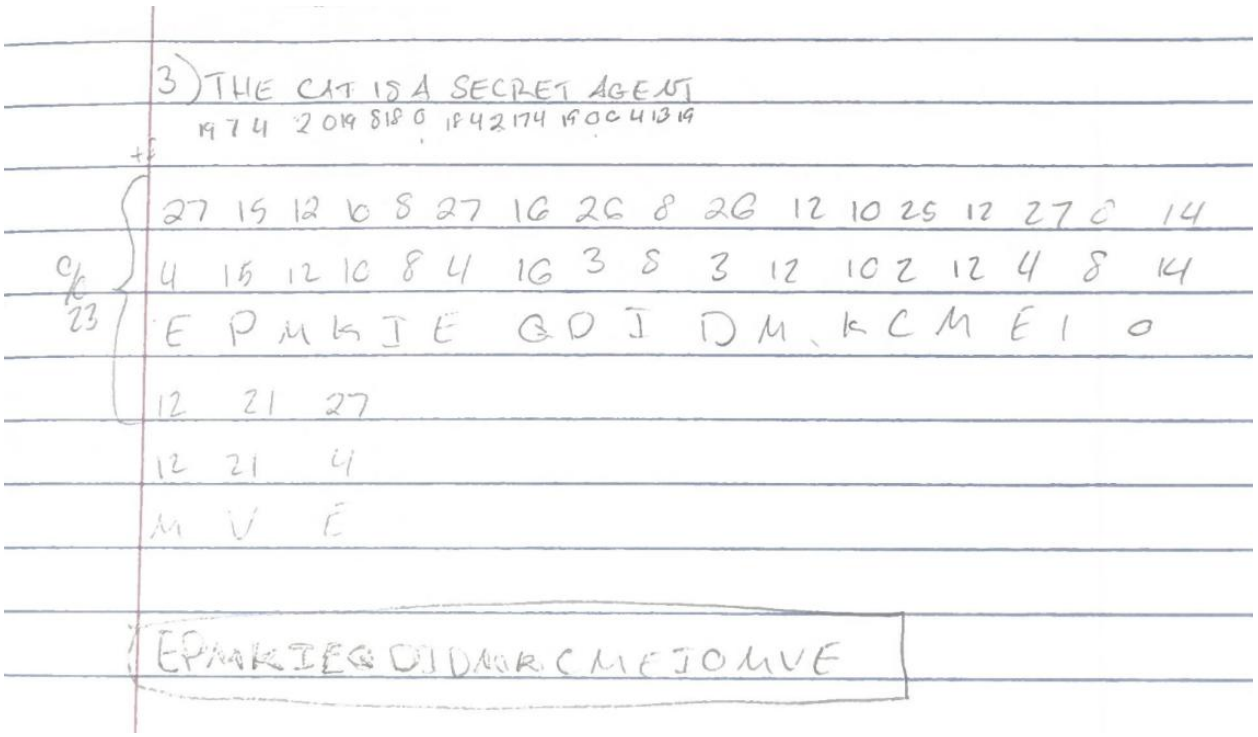
$= 100 \% 23$

$= 8$

$\boxed{\text{Shared key} = 8}$

3. Use the shared key as the rotation value (mod 23) to encrypt the message:

THE CAT IS A SECRET AGENT



4. Explain how B decrypts the message using the shared key.

   B decrypts the message by computing the same shard key using A's public key. Since the encryption added 8 to each letter's numeric value, B is able to decrypt by subtracting 8 from each ciphertext letter and applying mod 23. This is able to reverse the rotation and recover the original message.

Q2. **25 Points**

The following message was encrypted by user A using the ElGamal encryption algorithm.
Only uppercase letters are used with the mapping:
A = 0, B = 1, …, W = 22
(Letters X, Y, Z are not used)
Important: All calculations must be performed using mod 23 only. Do not use mod 26.

Message: REDCAT

Given:

- $p = 23, g = 5$
- A's private key: $x = 3$
- A selects a random value: $k = 2$

Questions

1. Using ElGamal, compute A's public key y.

1) $A_{pub} = 5^3 \% 23$

$= 125 \% 23$

$\boxed{A_{pub} = 10}$

2. Compute the value of $C1$.

$y = 10$

$y^k = 100$

$S = 100 \% 23 = 8$

$C_1 = g^k \% p$

2) $C_1 = 5^2 \% 23$

$= 25 \% 23$

$\boxed{= 2}$

3. A encrypts each letter of the message using ElGamal.

Write the ciphertext in the form:

$(C1, C2) = (_, (_, _, _, \ldots))$

3)	R	E	D	C	A	T
	17	4	3	2	0	19
* 8	136	32	24	16	0	152
% 23	21	9	1	16	0	14

$(C^1, C^2) = (2, (21, 9, 1, 16, 0, 14))$

1. Briefly explain how B would decrypt one ciphertext value to recover the original letter.

   To decrypt the ciphertext, B computer the shared secret. S = 8 (See document for calculation)

   B then computes the modular inverse. $s^{-1} = 3$.

   Each PT value is recovered using $m = C_2 * 3 \% 23$

Q3. **25 Points**

User A wants to send a secure safe opening code to user B. To ensure the code is authentic and has not been modified, A uses the ECDSA digital signature algorithm.

Shared Table (used by both A and B):

Position	Letter
1	G
2	H
3	K
4	M
5	P

6	R
7	S
8	T
9	U

Safe Code: 23456

A converts each digit of the safe code into a letter using the associated position in the shared table, and sends the resulting letters as the code.

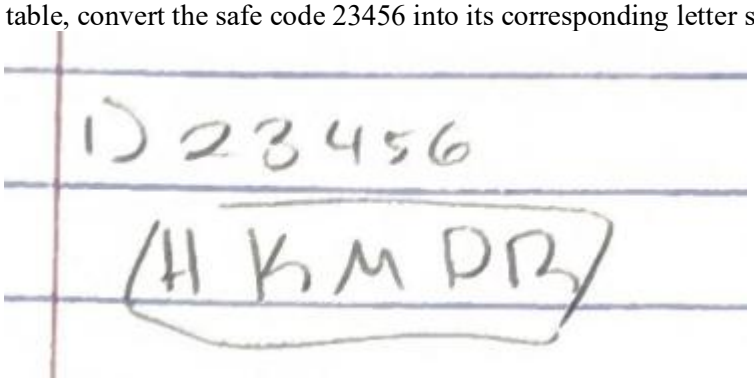
Given:

A's private key: d = 3

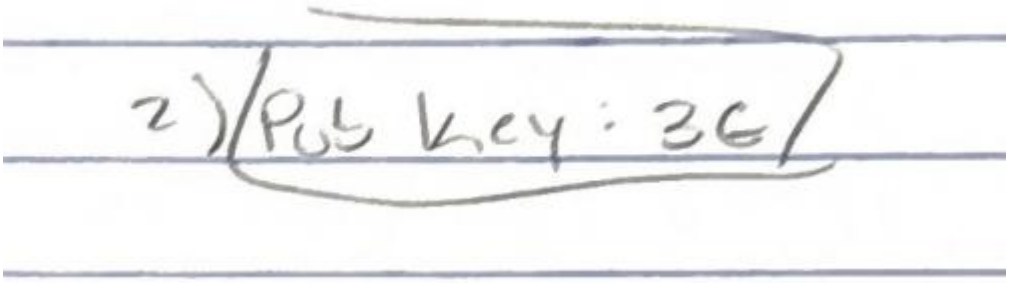
Base point: G

A selects a random value: k = 2

1. Using the shared table, convert the safe code 23456 into its corresponding letter sequence.



2. Using ECC, what is A's public key? (Public key = dG).



3. Using the given table, determine kG when k = 2:

3) kG when $k=2$

$kG = 2G$

$(=H)$

4. Convert your answer from Question 3 into its position in the table.

4) "H" is at position 2

5. Conceptually, what does the digital signature ensure about the safe code?

The digital signature ensures that the safe code is authentic and has not been modified, proving the code came from A and was not changed during the transmission.

6. Explain how B uses A's public key to verify the safe code.

B uses a digital signature to verify that the safe code came from A and was not altered during transmission. Using A's public key, B checks the signature against the received message and if it is verified, it proves that the code was generated by A's private key, the message maintained integrity during transmission, and is authentic.

Q4. Short Answer Questions 25 Points

a. In a digital signature system, a sender signs a message using their private key. What two security properties does a digital signature provide, and how are they achieved?

A digital signature provides authentication and integrity for a message. Authentication is achieved because only the sender's private key can generate the valid signature. Integrity because the signature would fail to authenticate if it were altered, this is achieved by using a hash of the message.

b. In the Diffie-Hellman key exchange, two users agree on public values $g$and p, and then exchange computed values over a public channel. Explain how both users are able to compute the same shared key without sending the key directly.

Each user only sends a public value that is found using their own private value (The exponent). Both users then apply their private value to the public value via an exponent, producing the same result: $(g^b)^a = (g^a)^b$

c. In the ElGamal encryption algorithm, ciphertext is represented as a pair $(C1, C2)$. Explain the role of $C1$and $C2$, and why a random value $k$is important in the encryption process.

C1 = g^k % p is the "randomness" that is needed for decryption
C2 = m (y^k) % p is the encrypted message
The random value k ensures that different ciphertexts are produced even when encrypting the same message twice.

d. Elliptic Curve Cryptography (ECC) is often preferred over RSA in modern systems. Explain one advantage of ECC over traditional public-key systems and why it is beneficial in practice.

ECC provides an equvalent security with smaller key sizses, meaning that there is faster computation, lower bandwith and resource usage, and better performance, especially on mobile devices.

e. In Elliptic Curve ElGamal encryption, scalar multiplication (e.g., kG) is used instead of exponentiation. Explain how ECC ElGamal differs from standard ElGamal and why elliptic curves are used instead of modular exponentiation.

Standard ElGamal uses modular exponentation over integers. ECC ElGamal however, replaces it with scalar multiplication on elliptic curve points. This provids the same security while keeping keys much smaller and manageable, resulting in better efficiency for ECC as well as improved security if it is improved upon which is even harder to break.